

Lab Title: Analyzing Monthly Sales Data for Central Tendencies

Problem Statement:

Understanding the central tendencies (mean, median, and mode) of monthly sales data can provide insights into business performance. By analyzing these metrics, businesses can identify trends and make informed decisions.

Objective:

- Compute the mean, median, and mode of monthly sales data.
- Interpret the significance of these measures in business analytics.

Dataset:

A sample dataset containing monthly sales figures:

Month	Sales (\$)
Jan	15000
Feb	18000
Mar	17000
Apr	16000
May	20000
Jun	19000
Jul	17000
Aug	21000
Sep	22000
Oct	18000
Nov	17000
Dec	20000

Requirements:

- Python installed (version 3.x recommended).
- NumPy and SciPy libraries installed (`pip install numpy scipy`).
- Jupyter Notebook (optional but recommended for running the lab).

Implementation:

```
In [1]: # Import necessary libraries
import numpy as np
```

```
from scipy import stats
```

```
In [2]: # Define monthly sales data
sales_data = np.array([15000, 18000, 17000, 16000, 20000, 19000, 17000, 21000, 2
```

```
In [3]: # Compute mean, median, and mode
mean_sales = np.mean(sales_data)
median_sales = np.median(sales_data)
mode_sales = stats.mode(sales_data, keepdims=True).mode[0]
```

```
In [4]: # Print results
print("Mean Sales:", mean_sales)
print("Median Sales:", median_sales)
print("Mode Sales:", mode_sales)
```

Mean Sales: 18333.333333333332

Median Sales: 18000.0

Mode Sales: 17000

```
In [ ]:
```